

Lab3-assignment-sentiment

March 3, 2025

1 Lab3 - Assignment Sentiment

Copyright: Vrije Universiteit Amsterdam, Faculty of Humanities, CLTL

This notebook describes the LAB-3 assignment of the Text Mining course. It is about sentiment analysis.

The aims of the assignment are: * Learn how to run a rule-based sentiment analysis module (VADER) * Learn how to run a machine learning sentiment analysis module (Scikit-Learn/ Naive Bayes) * Learn how to run scikit-learn metrics for the quantitative evaluation * Learn how to perform and interpret a quantitative evaluation of the outcomes of the tools (in terms of Precision, Recall, and F1) * Learn how to evaluate the results qualitatively (by examining the data) * Get insight into differences between the two applied methods * Get insight into the effects of using linguistic preprocessing * Be able to describe differences between the two methods in terms of their results * Get insight into issues when applying these methods across different domains

In this assignment, you are going to create your own gold standard set from 50 tweets. You will use the VADER and scikit-learn classifiers on these tweets and evaluate the results by using evaluation metrics and inspecting the data.

We recommend you go through the notebooks in the following order: * **Read the assignment (see below)** * **Lab3.2-Sentiment-analysis-with-VADER.ipynb** * **Lab3.3-Sentiment-analysis-with-scikit-learn.ipynb** * **Answer the questions of the assignment (see below) using the provided notebooks and submit**

In this assignment you are asked to perform both quantitative evaluations and error analyses: * a quantitative evaluation concerns the scores (Precision, Recall, and F1) provided by scikit's `classification_report`. It includes the scores per category, as well as micro and macro averages. Discuss whether the scores are balanced or not between the different categories (positive, negative, neutral) and between precision and recall. Discuss the shortcomings (if any) of the classifier based on these scores * an error analysis regarding the misclassifications of the classifier. It involves going through the texts and trying to understand what has gone wrong. It serves to get insight in what could be done to improve the performance of the classifier. Do you observe patterns in misclassifications? Discuss why these errors are made and propose ways to solve them.

1.1 Credits

The notebooks in this block have been originally created by [Marten Postma](#) and [Isa Maks](#). Adaptations were made by [Filip Ilievski](#).

1.2 Part I: VADER assignments

1.2.1 Preparation (nothing to submit):

To be able to answer the VADER questions you need to know how the tool works. * Read more about the VADER tool in [this blog](#).

* VADER provides 4 scores (positive, negative, neutral, compound). Be sure to understand what they mean and how they are calculated. * VADER uses rules to handle linguistic phenomena such as negation and intensification. Be sure to understand which rules are used, how they work, and why they are important. * VADER makes use of a sentiment lexicon. Have a look at the lexicon. Be sure to understand which information can be found there (lemma?, wordform?, part-of-speech?, polarity value?, word meaning?) What do all scores mean? https://github.com/cjhutto/vaderSentiment/blob/master/vaderSentiment/vader_lexicon.txt)

1.2.2 [3.5 points] Question1:

Regard the following sentences and their output as given by VADER. Regard sentences 1 to 7, and explain the outcome **for each sentence**. Take into account both the rules applied by VADER and the lexicon that is used. You will find that some of the results are reasonable, but others are not. Explain what is going wrong or not when correct and incorrect results are produced.

INPUT SENTENCE 1 I love apples

VADER OUTPUT {'neg': 0.0, 'neu': 0.192, 'pos': 0.808, 'compound': 0.6369}

INPUT SENTENCE 2 I don't love apples

VADER OUTPUT {'neg': 0.627, 'neu': 0.373, 'pos': 0.0, 'compound': -0.5216}

INPUT SENTENCE 3 I love apples :-)

VADER OUTPUT {'neg': 0.0, 'neu': 0.133, 'pos': 0.867, 'compound': 0.7579}

INPUT SENTENCE 4 These houses are ruins

VADER OUTPUT {'neg': 0.492, 'neu': 0.508, 'pos': 0.0, 'compound': -0.4404}

INPUT SENTENCE 5 These houses are certainly not considered ruins

VADER OUTPUT {'neg': 0.0, 'neu': 0.51, 'pos': 0.49, 'compound': 0.5867}

INPUT SENTENCE 6 He lies in the chair in the garden

VADER OUTPUT {'neg': 0.286, 'neu': 0.714, 'pos': 0.0, 'compound': -0.4215}

INPUT SENTENCE 7 This house is like any house

VADER OUTPUT {'neg': 0.0, 'neu': 0.667, 'pos': 0.333, 'compound': 0.3612}

1.3 Sentence Analysis

Overall notes about VADER: The VADER library is a very powerful and popular lexicon-based sentiment analysis tool used in many areas of natural language processing. VADER first breaks down the text into individual words(tokens), then each word(token) is assigned a sentiment score based on the “vader_lexicon.txt” file, identifying each word as either negative, neutral or positive. After all of this is done, VADER computes a ‘compound’ value which is the overall sentiment of the text. What makes VADER so powerful and popular is its ability to identify the intensity of

sentiment; for example, capitalization, punctuation, and reinforcing words (modifiers) have a large impact on the sentiment of the overall text.

Based on the notes about VADER, let's analyze the above sentences:

- Sentence 1: For this sentence the compound score seems reasonable, looking at each word in the sentence ['I', 'love', 'apples'] the word 'I' is mostly neutral, the word 'apples' does not exist within the lexicon file so is most likely assumed neutral as well, however; the word 'love' is in the lexicon file (and has an individual mean score of 3.2), accompanied by the fact that it is an intensifier for the neutral word 'I' the compound score makes sense here.
- Sentence 2: For this sentence the compound score seems more interesting, looking at each word again the only difference in this sentence is the addition of the word "don't," which negates the first sentence. Although the sentiment value for somebody writing 'I love apples' and 'I don't love apples' is the same for humans, VADER seems more unsure when the sentence is negated and therefore gives a greater 'neu' value. This shows us that common human text which includes jokes, irony, sarcasm, and negations will be difficult for VADER to understand due to its rule-based nature.
- Sentence 3: For this sentence the compound score makes a lot of sense, it is the same as the first sentence except for the addition of ':-)' at the end, in the 'vader_lexicon.txt' file the smiley face has a mean score of 1.3, and while it is not a hugely large score it does have some form of intensity, therefore, when coupled with 'love' brings the compound score from about 0.64 to 0.76 which is a significant difference.
- Sentence 4: For this sentence the compound score does not make a lot of sense, for humans such a sentence would have a significantly larger negative score than it does for VADER (depending on the context, of course). This sentence contains the word 'ruins' at the end (with a mean score of -1.9), and although there is heavy intensity projected onto the word 'houses,' VADER seems to not take this as much into account as humans would. This is another example of where complex language structures cannot be fully captured via lexicon(rule)-based systems.
- Sentence 5: For this sentence the compound does make sense, considering the explanation regarding sentence 4. Following the previous explanation, although a human would interpret this sentence more neutrally, the negation of the word 'ruins' and the inclusion of the word 'certainly' before the negation (with an individual mean score of 1.4). While the mean score for 'certainly' is not high because VADER struggles with the negation of the negative word 'ruins' and 'certainly' is a positive word (which can also be considered a slight intensifier), the compound score reflects these elements in the sentence.
- Sentence 6: For this sentence the compound score is very strange, for any human this sentence is purely informational, there is no emotion or response attached to this sentence without further context. Because the word 'lies' is present within the sentence (with an individual mean score of -1.8) and intensifies both 'chair' and 'garden,' the compound score reflects the intensity projected on the overall sentence.
- Sentence 7: Lastly, for this sentence the compound score is also a bit skewed, but less than sentence 6. Once again, this sentence is purely informational and meant to convey that a house is similar to other houses, although we are unsure of which qualities are compared, this is generally a neutral sentence. Having said that, we do also understand that because of the

word 'like' (which has an individual mean score of 1.5), and the intensity amplification on the word 'house,' that the compound score is slightly raised.

1.3.1 [Points: 2.5] Exercise 2: Collecting 50 tweets for evaluation

Collect 50 tweets. Try to find tweets that are interesting for sentiment analysis, e.g., very positive, neutral, and negative tweets. These could be your own tweets (typed in) or collected from the Twitter stream. If you have trouble accessing Twitter, try to find an existing dataset (on websites like kaggle or huggingface).

We will store the tweets in the file **my_tweets.json** (use a text editor to edit). For each tweet, you should insert: * sentiment analysis label: negative | neutral | positive (this you determine yourself, this is not done by a computer) * the text of the tweet * the Tweet-URL

from:

```
"1": {
    "sentiment_label": "",
    "text_of_tweet": "",
    "tweet_url": "",
```

to:

```
"1": {
    "sentiment_label": "positive",
    "text_of_tweet": "All across America people chose to get involved, get engaged and star",
    "tweet_url" : "https://twitter.com/BarackObama/status/946775615893655552",
},
```

You can load your tweets with human annotation in the following way.

```
[186]: import json
```

```
[187]: my_tweets = json.load(open('my_tweets.json'))
```

```
[188]: for id_, tweet_info in my_tweets.items():
        print(id_, tweet_info)
        break
```

```
1 {'sentiment_label': 'negative', 'text_of_tweet': 'michelle trachtenberg has
sadly passed away at the age of 39. rest in peace', 'tweet_url':
'https://x.com/PopCulture2000s/status/1894804536650334363'}
```

1.3.2 [5 points] Question 3:

Run VADER on your own tweets (see function **run_vader** from notebook **Lab2-Sentiment-analysis-using-VADER.ipynb**). You can use the code snippet below this explanation as a starting point. * [2.5 points] a. Perform a quantitative evaluation. Explain the different scores, and explain which scores are most relevant and why. * [2.5 points] b. Perform an error analysis: select 10 positive, 10 negative and 10 neutral tweets that are not correctly classified and try to understand why. Refer to the VADER-rules and the VADER-lexicon. Of course, if there are less than 10 errors

for a category, you only have to check those. For example, if there are only 5 errors for positive tweets, you just describe those.

```
[189]: from nltk.sentiment import SentimentIntensityAnalyzer
from sklearn.metrics import classification_report

sia = SentimentIntensityAnalyzer()

def run_vader(tweet):
    """
    Runs VADER sentiment analysis on a given tweet.

    :param str tweet: The text of the tweet
    :return: The VADER sentiment output as a dictionary
    """
    return sia.polarity_scores(tweet)

def vader_output_to_label(vader_output):
    """
    map vader output e.g.,
    {'neg': 0.0, 'neu': 0.0, 'pos': 1.0, 'compound': 0.4215}
    to one of the following values:
    a) positive float -> 'positive'
    b) 0.0 -> 'neutral'
    c) negative float -> 'negative'

    :param dict vader_output: output dict from vader

    :rtype: str
    :return: 'negative' | 'neutral' | 'positive'
    """
    compound = vader_output['compound']

    if compound < 0:
        return 'negative'
    elif compound == 0.0:
        return 'neutral'
    elif compound > 0.0:
        return 'positive'

assert vader_output_to_label( {'neg': 0.0, 'neu': 0.0, 'pos': 1.0, 'compound': 0.0}) == 'neutral'
assert vader_output_to_label( {'neg': 0.0, 'neu': 0.0, 'pos': 1.0, 'compound': 0.01}) == 'positive'
assert vader_output_to_label( {'neg': 0.0, 'neu': 0.0, 'pos': 1.0, 'compound': -0.01}) == 'negative'
```

```
[190]: tweets = []
all_vader_output = []
gold = []

# settings (to change for different experiments)
to_lemmatize = True
pos = set()

for id_, tweet_info in my_tweets.items():
    the_tweet = tweet_info['text_of_tweet']
    vader_output = run_vader(the_tweet) # run vader
    vader_label = vader_output_to_label(vader_output) # convert vader output to
    ↪ category

    tweets.append(the_tweet)
    all_vader_output.append(vader_label)
    gold.append(tweet_info['sentiment_label'])

# use scikit-learn's classification report
print(classification_report(gold, all_vader_output))
```

	precision	recall	f1-score	support
negative	0.67	0.78	0.72	18
neutral	0.86	0.35	0.50	17
positive	0.59	0.87	0.70	15
accuracy			0.66	50
macro avg	0.70	0.67	0.64	50
weighted avg	0.71	0.66	0.64	50

2 Explanation of relevant scores and why they are relevant

Our overall interpretation: * Neutral tweets have the highest precision (0.86), meaning that when VADER classifies a tweet as neutral, it is usually correct. However, since its recall for neutral tweets is very low (0.35), it often fails to identify neutral tweets in the first place. Negative tweets have a precision of 0.67, showing that while VADER performs reasonably well, it still classifies some tweets as negative when they are not. Positive tweets have the lowest precision (0.59), indicating that VADER frequently overestimates positivity. VADER tends to misclassify tweets as positive more often than negative or neutral.

- Positive tweets have a high recall (0.87), meaning it captures most positive tweets but at the cost of lower precision as previously mentioned. Negative tweets also have a high recall (0.78), which shows that VADER successfully detects most negative tweets. However, neutral tweets have very low recall (0.35), indicating that VADER struggles to recognize them and often wrongly classifies them as positive or negative. This suggests VADER is better at detecting strong sentiments but struggles with more neutral tweets.

- Negative tweets have an F1-score of 0.72, meaning VADER performs fairly well in identifying them, as do positive tweets with a similar F1-score of 0.70. Neutral tweets have the lowest F1-score (0.50), reflecting VADER's difficulty in correctly identifying them due to a low recall score. This suggests that while VADER handles strong sentiments relatively well, it struggles the most with neutral tweets.
- The support scores indicate the number of tweets in each sentiment category. There are 18 negative, 17 neutral, and 15 positive tweets, meaning the dataset is relatively balanced but has a slight favor for negative tweets.

[191]: `print(tweets)`

```
['michelle trachtenberg has sadly passed away at the age of 39. rest in peace',
'Your a War Criminal - Thousands are dead because of you', 'I guess this is a
controversial opinion but men can be raped and abused by women', "If Elon didn't
buy Twitter, it's entirely possible that Kamala would be president right now.
Terrifying thought", 'Leaking ICE raid details is not just betrayal, it is a
national security risk', 'Nazis and racists get to spew hate all day on this app
but NBA Centel does a little trolling and gets banned. interesting', 'Dad
looking annoyed and mum crying with disappointment that it's a girl', 'As a
father this has to be your worst nightmare', "Complete batshit lunacy on the
president's Truth Social account", 'Nigga that got killed by that white lady
digital footprint is HORRIBLE', 'Zookeepers in China gave this poor lion a bad
haircut. Faced Public Backlash.', 'I would be so pissed if I lived in lumiose
city', 'every animal i've interacted with today has been very mean to me and it
makes me upset', 'I am genuinely losing my mind what in the ever-loving hell is
the minecraft movie', 'Federal worker starts crying about losing her job', 'So
fascinated by these people who think seed oils and vaccines are dangerous and
unnatural but still get filler', 'On Facebook the bastard who owns it needs to
stop all the animal cruelty videos,in fact any animal videos it is extremely
cruel violent abuse of innocent animals the fuckers abusing monkeys need
tracking down and dealing with and all who think it ok', 'The most dangerous
animal on the planet, a violent creature filled with hateful thoughts and a lust
for blood', 'I'm convinced that pandas are just drunk people in costumes',
'Animals reacting to themselves in the mirror in jungle', 'Penguin paints a
picture and is extremely pleased with herself', 'The professor banned laptops so
the students had to find a way', 'Unfortunately, as President Bukele eloquently
articulates, there is no other option', 'On this day one year ago, TikToker
mr.weatherall posted the following video', 'rest in peace michelle trachtenberg.
thank you for the childhood memories', 'I regret to inform you that this site
still has the juice', 'This is a very important dog. This dog has a good
heart.', 'Fire department and animal control personnel rescue a very scared and
cold dog', 'Elephant throwing dirt on a Nile Crocodile for some reason', 'Elon
Musk has been receiving many death threats lately because he is exposing deep-
rooted corruption & working to save the country from bankruptcy-all without
charging a single penny', 'An elderly Lion in his final hours', 'Elon Musk has
surpassed 219 million followers. He is the most followed account on ', '9
months old baby horse sees sea for the first time', 'this is so unfunny yet the
funniest thing ever to me', 'All citizens of Sweden are Swedish', 'The Black
```

Panther closed his eyes & just seemed so content. My heart melted ', 'Rocket, the majestic male tiger of the Pilibhit Tiger Reserve, in India truly lives up to his bold and regal appearance', 'This is so beautiful it's almost scary', 'A beautiful bond between K-9 and his handler', 'I'll say, it's nice to have someone actually say this', 'Thirty-two years together and you still take my breath away. Happy Valentine's Day', '10 terrifying beauties of wildlife', 'Gorgeous grey wolf becomes a good boy when visited by the people that helped raise their pack', 'THIS DOG AND DEER BEING FRIENDS IS LITERALLY THE CUTEST THING EVER ', 'I have no idea what is happening here but it is the best thing I've seen today', 'This is my favorite video on the internet', 'The swans love dance in winter', 'The best shot in television history', 'I'm proud to be Swedish because of my ancestors' hard work in building my beloved Sweden', 'Beautiful bonding between Mother-Son']

```
[192]: print(all_vader_output)
```

```
['positive', 'negative', 'negative', 'negative', 'positive', 'negative',  
'negative', 'negative', 'positive', 'negative', 'negative', 'negative',  
'negative', 'negative', 'negative', 'positive', 'negative', 'negative',  
'positive', 'neutral', 'positive', 'negative', 'negative', 'neutral',  
'positive', 'negative', 'positive', 'negative', 'negative', 'negative',  
'neutral', 'neutral', 'neutral', 'positive', 'neutral', 'neutral', 'positive',  
'positive', 'positive', 'positive', 'positive', 'negative', 'positive',  
'positive', 'positive', 'positive', 'positive', 'positive', 'positive',  
'positive']
```

```
[193]: print(gold)
```

```
['negative', 'negative', 'negative', 'negative', 'negative', 'negative',  
'negative', 'negative', 'negative', 'negative', 'negative', 'negative',  
'negative', 'negative', 'negative', 'negative', 'negative', 'negative',  
'neutral', 'neutral', 'neutral', 'neutral', 'neutral', 'neutral', 'neutral',  
'neutral', 'neutral', 'neutral', 'neutral', 'neutral', 'neutral', 'neutral',  
'neutral', 'neutral', 'neutral', 'positive', 'positive', 'positive', 'positive',  
'positive', 'positive', 'positive', 'positive', 'positive', 'positive',  
'positive', 'positive', 'positive', 'positive', 'positive', 'positive']
```

```
[194]: incorrect_positive = []  
incorrect_negative = []  
incorrect_neutral = []  
  
for tweet, pred, actual in zip(tweets, all_vader_output, gold):  
    if pred != actual:  
        if actual == "positive":  
            incorrect_positive.append(tweet)  
        elif actual == "negative":  
            incorrect_negative.append(tweet)  
        elif actual == "neutral":
```



```

        incorrect_neutral.append(tweet)

# Show misclassified tweets (up to 10 per category)
print("Incorrectly classified positive tweets:", incorrect_positive[:10],
      ↪len(incorrect_positive), 'tweets')
print()
print("Incorrectly classified negative tweets:", incorrect_negative[:10],
      ↪len(incorrect_negative), 'tweets')
print()
print("Incorrectly classified neutral tweets:", incorrect_neutral[:10],
      ↪len(incorrect_neutral), 'tweets')

```

Incorrectly classified positive tweets: ['The Black Panther closed his eyes & just seemed so content. My heart melted ', '10 terrifying beauties of wildlife'] 2 tweets

Incorrectly classified negative tweets: ['michelle trachtenberg has sadly passed away at the age of 39. rest in peace', 'Leaking ICE raid details is not just betrayal, it is a national security risk', 'Complete batshit lunacy on the president's Truth Social account', 'So fascinated by these people who think seed oils and vaccines are dangerous and unnatural but still get filler'] 4 tweets

Incorrectly classified neutral tweets: ['I'm convinced that pandas are just drunk people in costumes', 'Penguin paints a picture and is extremely pleased with herself', 'The professor banned laptops so the students had to find a way', 'Unfortunately, as President Bukele eloquently articulates, there is no other option', 'rest in peace michelle trachtenberg. thank you for the childhood memories', 'I regret to inform you that this site still has the juice', 'This is a very important dog. This dog has a good heart.', 'Fire department and animal control personnel rescue a very scared and cold dog', 'Elephant throwing dirt on a Nile Crocodile for some reason', 'Elon Musk has been receiving many death threats lately because he is exposing deep-rooted corruption & working to save the country from bankruptcy-all without charging a single penny'] 11 tweets

3 Error analysis and explanation

The error analysis identifies the misclassified positive, negative, and neutral tweets.

- There are 2 misclassified positive tweets identified meaning they were originally labelled as positive but VADER failed to do so. For example, the sentence: “The Black Panther closed his eyes & just seemed so content. My heart melted ” is misclassified as either a neutral or negative tweet. VADER recognizes words like “heart” and “content” not as positive cues because they are not explicitly strong enough to be recognized as positive words. In the sentence “10 terrifying beauties of wildlife” VADER probably misclassified since it contains the word “terrifying”, even if it is intended as a positive sentence.
- There are 4 misclassified negative tweets identified meaning they were originally labelled as negative but VADER failed to do so. For example, the sentence: “Complete batshit lunacy on the president’s Truth Social account” is misclassified as either a neutral or positive tweet.

VADER doesn't recognize informal and rather humorous words like "batshit" and "lunacy".

- There are 11 misclassified neutral tweets identified meaning they were originally labelled as neutral but VADER failed to do so. For example, the sentence: "I'm convinced that pandas are just drunk people in costumes" is misclassified as either a positive or negative tweet. This could be because VADER interpreted the word "drunk" as negative and has problems with identifying sentences that are meant as funny which humans would mostly label as neutral.

3.0.1 [4 points] Question 4:

Run VADER on the set of airline tweets with the following settings:

- Run VADER (as it is) on the set of airline tweets
- Run VADER on the set of airline tweets after having lemmatized the text
- Run VADER on the set of airline tweets with only adjectives
- Run VADER on the set of airline tweets with only adjectives and after having lemmatized the text
- Run VADER on the set of airline tweets with only nouns
- Run VADER on the set of airline tweets with only nouns and after having lemmatized the text
- Run VADER on the set of airline tweets with only verbs
- Run VADER on the set of airline tweets with only verbs and after having lemmatized the text
- [1 point] a. Generate for all separate experiments the classification report, i.e., Precision, Recall, and F1 scores per category as well as micro and macro averages. **Use a different code cell (or multiple code cells) for each experiment.**
- [3 points] b. Compare the scores and explain what they tell you.
 - Does lemmatisation help? Explain why or why not.
 - Are all parts of speech equally important for sentiment analysis? Explain why or why not.

```
[198]: # Your code here
from sklearn.datasets import load_files
from collections import Counter
import spacy
import pathlib

nlp = spacy.load('en_core_web_sm')
cwd = pathlib.Path.cwd()
airline_tweets_folder = cwd.joinpath('airlinetweets')
print('path:', airline_tweets_folder)

def run_vader(textual_unit,
              lemmatize=False,
```

```

        parts_of_speech_to_consider=None,
        verbose=0):
    """
    Run VADER on a sentence from spacy

    :param str textual_unit: a textual unit, e.g., sentence, sentences (one_
    ↪string)
    (by looping over doc.sents)
    :param bool lemmatize: If True, provide lemmas to VADER instead of words
    :param set parts_of_speech_to_consider:
    -None or empty set: all parts of speech are provided
    -non-empty set: only these parts of speech are considered.
    :param int verbose: if set to 1, information is printed
    about input and output

    :rtype: dict
    :return: vader output dict
    """
    doc = nlp(textual_unit)

    input_to_vader = []

    for sent in doc.sents:
        for token in sent:

            to_add = token.text

            if lemmatize:
                to_add = token.lemma_

                if to_add == '-PRON-':
                    to_add = token.text

            if parts_of_speech_to_consider:
                if token.pos_ in parts_of_speech_to_consider:
                    input_to_vader.append(to_add)
            else:
                input_to_vader.append(to_add)

    scores = sia.polarity_scores(' '.join(input_to_vader))

    if verbose >= 1:
        print()
        print('INPUT SENTENCE', sent)
        print('INPUT TO VADER', input_to_vader)
        print('VADER OUTPUT', scores)

```

```
return scores
```

```
path: /Users/robertostoica/Desktop/University_files_(year_3)_extra/period_4/Text Mining for AI/Assignments/Lab_Assignments/Lab_3/airlinetweets
```

```
[199]: from sklearn import metrics
path = str(airline_tweets_folder)
airline_tweets_train = load_files(path)
print(airline_tweets_train.target_names)

freqs = Counter(airline_tweets_train.target)
print(freqs)
print()
for category, frequency in freqs.items():
    print(category, frequency)
    print(airline_tweets_train.target_names[category], frequency)
```

```
['negative', 'neutral', 'positive']
Counter({0: 1750, 1: 1515, 2: 1490})
```

```
1 1515
neutral 1515
2 1490
positive 1490
0 1750
negative 1750
```

```
[201]: airline_tweets_train = load_files(airline_tweets_folder, encoding="utf-8")

# Define different configurations
cases = {
    "Baseline VADER": {"lemmatize": False, "parts_of_speech_to_consider": None},
    "Lemmatized VADER": {"lemmatize": True, "parts_of_speech_to_consider": None},
    "VADER with only adjectives": {"lemmatize": False,
    "parts_of_speech_to_consider": {"ADJ"}},
    "VADER with only adjectives + lemmatization": {"lemmatize": True,
    "parts_of_speech_to_consider": {"ADJ"}},
    "VADER with only nouns": {"lemmatize": False, "parts_of_speech_to_consider": {"NOUN"}},
    "VADER with only nouns + lemmatization": {"lemmatize": True,
    "parts_of_speech_to_consider": {"NOUN"}},
    "VADER with only verbs": {"lemmatize": False, "parts_of_speech_to_consider": {"VERB"}},
    "VADER with only verbs + lemmatization": {"lemmatize": True,
    "parts_of_speech_to_consider": {"VERB"}},
}
```

```

# Store classification reports
reports = {}

# Assuming airline_tweets_train has 'target' as ground truth sentiment
↳categories
true_labels = [airline_tweets_train.target_names[label] for label in
↳airline_tweets_train.target]
    # Adjust based on dataset structure

# Loop through each case
for case_name, params in cases.items():
    baseline_text_lst = []
    predictions = []

    for sentence in airline_tweets_train.data: # No need to decode
        baseline_output = run_vader(sentence, **params)

        # Store processed text and its sentiment output
        baseline_text_lst.append((sentence, baseline_output))

        # Convert VADER compound score to sentiment category
        compound_score = baseline_output["compound"]
        if compound_score >= 0.05:
            predictions.append("positive")
        elif compound_score <= -0.05:
            predictions.append("negative")
        else:
            predictions.append("neutral")

    # Generate classification report
    reports[case_name] = classification_report(true_labels, predictions,
↳target_names=["negative", "neutral", "positive"])

# Print results
print(f"==== {case_name} ====")
print(reports[case_name])
print("\n" + "="*50 + "\n")

```

==== Baseline VADER ====

	precision	recall	f1-score	support
negative	0.80	0.51	0.62	1750
neutral	0.60	0.51	0.55	1515
positive	0.56	0.88	0.69	1490
accuracy			0.63	4755

macro avg	0.65	0.64	0.62	4755
weighted avg	0.66	0.63	0.62	4755

=====

==== Lemmatized VADER ====

	precision	recall	f1-score	support
negative	0.79	0.52	0.62	1750
neutral	0.58	0.49	0.54	1515
positive	0.56	0.88	0.69	1490
accuracy			0.62	4755
macro avg	0.64	0.63	0.61	4755
weighted avg	0.65	0.62	0.61	4755

=====

==== VADER with only adjectives ====

	precision	recall	f1-score	support
negative	0.86	0.20	0.33	1750
neutral	0.40	0.89	0.55	1515
positive	0.67	0.44	0.53	1490
accuracy			0.50	4755
macro avg	0.64	0.51	0.47	4755
weighted avg	0.65	0.50	0.46	4755

=====

==== VADER with only adjectives + lemmatization ====

	precision	recall	f1-score	support
negative	0.86	0.20	0.33	1750
neutral	0.40	0.89	0.55	1515
positive	0.67	0.44	0.53	1490
accuracy			0.50	4755
macro avg	0.64	0.51	0.47	4755
weighted avg	0.65	0.50	0.46	4755

=====

==== VADER with only nouns ====

	precision	recall	f1-score	support
negative	0.73	0.14	0.23	1750
neutral	0.36	0.82	0.50	1515
positive	0.54	0.35	0.42	1490
accuracy			0.42	4755
macro avg	0.54	0.44	0.39	4755
weighted avg	0.55	0.42	0.38	4755

=====

==== VADER with only nouns + lemmatization ====

	precision	recall	f1-score	support
negative	0.72	0.15	0.25	1750
neutral	0.36	0.81	0.50	1515
positive	0.53	0.34	0.41	1490
accuracy			0.42	4755
macro avg	0.53	0.44	0.39	4755
weighted avg	0.54	0.42	0.38	4755

=====

==== VADER with only verbs ====

	precision	recall	f1-score	support
negative	0.79	0.28	0.42	1750
neutral	0.38	0.81	0.52	1515
positive	0.58	0.35	0.43	1490
accuracy			0.47	4755
macro avg	0.58	0.48	0.46	4755
weighted avg	0.59	0.47	0.46	4755

=====

==== VADER with only verbs + lemmatization ====

	precision	recall	f1-score	support
negative	0.75	0.29	0.42	1750
neutral	0.38	0.78	0.51	1515
positive	0.58	0.36	0.44	1490

accuracy			0.47	4755
macro avg	0.57	0.48	0.46	4755
weighted avg	0.58	0.47	0.46	4755

=====

3.1 Analysis of results

After examining the scores, we can say the following: The baseline run of VADER produced the best output scores from all the different experiments we ran, however, if the data is too large, running VADER on the text only with verbs produces a decent enough result to use in a large data pipeline. However, due to the simple nature of this assignment it makes sense that VADER needs as much text as possible to get the best scores possible.

After examining the results, we can say that lemmatization does not seem to help VADER when analyzing the airline tweets. This is most likely because the vocabulary used within tweets is already very basic and condensed, due to this property of tweets, it means that lemmatizing the text produces a similar output to the original text of the tweet. Therefore, not having a significant impact on the results.

The parts of speech have different importance in the analysis that VADER does. VADER does not take into account neutral words which do not have any intensifying negative/positive word within the sentence. This means that most of the words sentences are simply neutral and not considered unless directly acted upon by a word with sentiment value within the provided context. For this simple reason, not all parts of speech are equally important for sentiment analysis.

3.2 We also provide the raw outputs as a reference to see how the different parameters affect the output

[159]: *# Running VADER as is on the set of airline tweets*

```
baseline_text_lst = []
for sentence in airline_tweets_train.data:
    baseline_text = sentence.decode("utf-8")
    baseline_output = run_vader(baseline_text, verbose=0)
    baseline_text_lst.append((baseline_text, baseline_output))
print(baseline_text_lst[0:30])
```

```
[('@AmericanAir Why is your cover photo of TWA? Just wondering.', {'neg': 0.0,
'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('@united next day', {'neg': 0.0,
'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('"@JetBlue Thank you that it is not
just a livery; it is a culture that 16', {'neg': 0.0, 'neu': 0.839, 'pos':
0.161, 'compound': 0.3612}), ('@SouthwestAir I cannot DM you as you do not
follow me.', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}),
('"@VirginAmerica Funny story', {'neg': 0.0, 'neu': 0.408, 'pos': 0.592,
'compound': 0.4404}), ('@united flight 403 is delayed 40 min bc a missing screw
```


and were in danger of missing flight 6491', {'neg': 0.461, 'neu': 0.539, 'pos': 0.0, 'compound': -0.8442}), ('@AmericanAir does it show that we are on the same plane?', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('@united @perfectomobile You really shouldn't use the word 'crashing.' Just sayin'.", {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('@united what's a good number to call to speak with someone about how you can fix what you did to 50 people and their luggage on Saturday?", {'neg': 0.0, 'neu': 0.856, 'pos': 0.144, 'compound': 0.4939}), ('@USAirways Im researching a flight & noticed a small issue w/class of service between your site & other sites. Could Reservations help me?', {'neg': 0.0, 'neu': 0.876, 'pos': 0.124, 'compound': 0.4019}), ('"@united never fails', {'neg': 0.0, 'neu': 0.462, 'pos': 0.538, 'compound': 0.3252}), ('"@united hi guys', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('"@united broken light', {'neg': 0.608, 'neu': 0.392, 'pos': 0.0, 'compound': -0.4767}), ('@united to speak to a real person to get this resolved politely and efficiently. (2/2)', {'neg': 0.0, 'neu': 0.724, 'pos': 0.276, 'compound': 0.5537}), ('@united love the new 1st class breakfast!', {'neg': 0.0, 'neu': 0.572, 'pos': 0.428, 'compound': 0.6696}), ('"@USAirways just heard #Monterey airport was closed due to thick fog', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('@united I did DM the details but still no response. I have to book today.', {'neg': 0.189, 'neu': 0.811, 'pos': 0.0, 'compound': -0.4215}), ('@united yes 1427 Cancelled Flightled. Moved to 333. I'll figure out car rental changes and such.", {'neg': 0.107, 'neu': 0.749, 'pos': 0.144, 'compound': 0.1779}), ('"@USAirways awesome! And yes', {'neg': 0.0, 'neu': 0.22, 'pos': 0.78, 'compound': 0.7959}), ('@united it was UA flight 1001! Now I'm currently stuck in Portland because I missed my ride because of the 3 hour delay. #GetMartyHome", {'neg': 0.263, 'neu': 0.737, 'pos': 0.0, 'compound': -0.6996}), ('@SouthwestAir nice touch on the passbook integration!', {'neg': 0.0, 'neu': 0.66, 'pos': 0.34, 'compound': 0.4753}), ('Keep it up :) @AmericanAir', {'neg': 0.0, 'neu': 0.571, 'pos': 0.429, 'compound': 0.4588}), ('@AmericanAir DONE!! thank you so much!!', {'neg': 0.0, 'neu': 0.577, 'pos': 0.423, 'compound': 0.5673}), ('@USAirways any chance of a travel advisory or DCA this weekend?', {'neg': 0.0, 'neu': 0.818, 'pos': 0.182, 'compound': 0.25}), ('@JetBlue keep going back and forth with being able to board and then not. But great agents in savannah!', {'neg': 0.208, 'neu': 0.792, 'pos': 0.0, 'compound': -0.694}), ('"@JetBlue Flight 152 (N559JB) ""Here\'s Looking at Blue Kid"" departs @MCO enroute to @BostonLogan <http://t.co/qeaDA92MW6>", {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('"@united diverged to Burlington', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('@united thank you for taking care of my mom and reFlight Booking Problems her flight in phl.', {'neg': 0.121, 'neu': 0.625, 'pos': 0.254, 'compound': 0.4588}), ('@united thanks we'll be in touch!", {'neg': 0.0, 'neu': 0.653, 'pos': 0.347, 'compound': 0.4926}), ('@united line full of worried customers and guy at desk says so sorry you will have to take care of your problems Late Flightr I don't have time.", {'neg': 0.196, 'neu': 0.709, 'pos': 0.095, 'compound': -0.3488})]

[161]: *# Running VADER on the lemmas of the text*

```

lemmatized_text_lst = []
for sentence in airline_tweets_train.data:
    lemmatized_text = sentence.decode("utf-8")
    lemmatized_output = run_vader(lemmatized_text, lemmatize=True, verbose=0)
    lemmatized_text_lst.append((lemmatized_text, lemmatized_output))
print(lemmatized_text_lst[0:30])

```

```

[('@AmericanAir Why is your cover photo of TWA? Just wondering.', {'neg': 0.0,
'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('@united next day', {'neg': 0.0,
'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('"@JetBlue Thank you that it is not
just a livery; it is a culture that 16', {'neg': 0.0, 'neu': 0.839, 'pos':
0.161, 'compound': 0.3612}), ('@SouthwestAir I cannot DM you as you do not
follow me.', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}),
('"@VirginAmerica Funny story', {'neg': 0.0, 'neu': 0.408, 'pos': 0.592,
'compound': 0.4404}), ('@united flight 403 is delayed 40 min bc a missing screw
and were in danger of missing flight 6491', {'neg': 0.442, 'neu': 0.558, 'pos':
0.0, 'compound': -0.8074}), ('@AmericanAir does it show that we are on the same
plane?', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('@united
@perfectomobile You really shouldn't use the word 'crashing.' Just sayin'.",
{'neg': 0.197, 'neu': 0.803, 'pos': 0.0, 'compound': -0.4019}), ('@united what's
a good number to call to speak with someone about how you can fix what you did
to 50 people and their luggage on Saturday?", {'neg': 0.0, 'neu': 0.856, 'pos':
0.144, 'compound': 0.4939}), ('@USAirways Im researching a flight & noticed
a small issue w/class of service between your site & other sites. Could
Reservations help me?', {'neg': 0.0, 'neu': 0.87, 'pos': 0.13, 'compound':
0.4019}), ('"@united never fails', {'neg': 0.0, 'neu': 0.412, 'pos': 0.588,
'compound': 0.431}), ('"@united hi guys', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0,
'compound': 0.0}), ('"@united broken light', {'neg': 0.0, 'neu': 1.0, 'pos':
0.0, 'compound': 0.0}), ('@united to speak to a real person to get this resolved
politely and efficiently. (2/2)', {'neg': 0.0, 'neu': 0.678, 'pos': 0.322,
'compound': 0.6908}), ('@united love the new 1st class breakfast!', {'neg': 0.0,
'neu': 0.572, 'pos': 0.428, 'compound': 0.6696}), ('"@USAirways just heard
#Monterey airport was closed due to thick fog', {'neg': 0.0, 'neu': 1.0, 'pos':
0.0, 'compound': 0.0}), ('@united I did DM the details but still no response. I
have to book today.', {'neg': 0.189, 'neu': 0.811, 'pos': 0.0, 'compound':
-0.4215}), ('@united yes 1427 Cancelled Flightled. Moved to 333. I'll figure out
car rental changes and such.", {'neg': 0.107, 'neu': 0.749, 'pos': 0.144,
'compound': 0.1779}), ('"@USAirways awesome! And yes', {'neg': 0.0, 'neu': 0.22,
'pos': 0.78, 'compound': 0.7959}), ('@united it was UA flight 1001! Now I'm
currently stuck in Portland because I missed my ride because of the 3 hour
delay. #GetMartyHome", {'neg': 0.173, 'neu': 0.827, 'pos': 0.0, 'compound':
-0.4926}), ('@SouthwestAir nice touch on the passbook integration!', {'neg':
0.0, 'neu': 0.66, 'pos': 0.34, 'compound': 0.4753}), ('Keep it up :)
@AmericanAir', {'neg': 0.0, 'neu': 0.571, 'pos': 0.429, 'compound': 0.4588}),
('@AmericanAir DONE!! thank you so much!!', {'neg': 0.0, 'neu': 0.577, 'pos':
0.423, 'compound': 0.5673}), ('@USAirways any chance of a travel advisory or DCA
this weekend?', {'neg': 0.0, 'neu': 0.818, 'pos': 0.182, 'compound': 0.25}),
('@JetBlue keep going back and forth with being able to board and then not. But

```

```
great agents in savannah!', {'neg': 0.208, 'neu': 0.792, 'pos': 0.0, 'compound':
-0.694}), ('"@JetBlue Flight 152 (N559JB) ""Here\'s Looking at Blue Kid""
departs @MCO enroute to @BostonLogan http://t.co/qeaDA92MW6"', {'neg': 0.0,
'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('"@united diverged to Burlington',
{'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('@united thank you for
taking care of my mom and reFlight Booking Problems her flight in phl.', {'neg':
0.121, 'neu': 0.625, 'pos': 0.254, 'compound': 0.4588}), ('@united thanks we'll
be in touch!", {'neg': 0.0, 'neu': 0.682, 'pos': 0.318, 'compound': 0.4199}),
('"@united line full of worried customers and guy at desk says so sorry you will
have to take care of your problems Late Flightr I don't have time.", {'neg':
0.196, 'neu': 0.709, 'pos': 0.095, 'compound': -0.3488})]
```

[166]: *# Running VADER searching for the adjectives*

```
adjective_text_lst = []

for sentence in airline_tweets_train.data:
    adjective_text = sentence.decode("utf-8")
    adjective_output = run_vader(adjective_text, lemmatize=False,
↳ parts_of_speech_to_consider={'JJ'}, verbose=0)
    adjective_text_lst.append((adjective_text, adjective_output))
print(adjective_text_lst[0:30])
```

```
[('@AmericanAir Why is your cover photo of TWA? Just wondering.', {'neg': 0.0,
'neu': 0.0, 'pos': 0.0, 'compound': 0.0}), ('@united next day', {'neg': 0.0,
'neu': 0.0, 'pos': 0.0, 'compound': 0.0}), ('"@JetBlue Thank you that it is not
just a livery; it is a culture that 16', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0,
'compound': 0.0}), ('@SouthwestAir I cannot DM you as you do not follow me.',
{'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0}), ('"@VirginAmerica Funny
story', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0}), ('@united flight
403 is delayed 40 min bc a missing screw and were in danger of missing flight
6491', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0}), ('@AmericanAir
does it show that we are on the same plane?', {'neg': 0.0, 'neu': 0.0, 'pos':
0.0, 'compound': 0.0}), ('@united @perfectomobile You really shouldn't use the
word 'crashing.' Just sayin'.', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound':
0.0}), ('@united what's a good number to call to speak with someone about how
you can fix what you did to 50 people and their luggage on Saturday?', {'neg':
0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0}), ('@USAirways Im researching a
flight & noticed a small issue w/class of service between your site &
other sites. Could Reservations help me?', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0,
'compound': 0.0}), ('"@united never fails', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0,
'compound': 0.0}), ('"@united hi guys', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0,
'compound': 0.0}), ('"@united broken light', {'neg': 0.0, 'neu': 0.0, 'pos':
0.0, 'compound': 0.0}), ('@united to speak to a real person to get this resolved
politely and efficiently. (2/2)', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0,
'compound': 0.0}), ('@united love the new 1st class breakfast!', {'neg': 0.0,
'neu': 0.0, 'pos': 0.0, 'compound': 0.0}), ('"@USAirways just heard #Monterey
airport was closed due to thick fog', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0,
```

```
'compound': 0.0})), ('@united I did DM the details but still no response. I have
to book today.', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0})),
("@united yes 1427 Cancelled Flightled. Moved to 333. I'll figure out car rental
changes and such.", {'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0})),
('"@USAirways awesome! And yes', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0,
'compound': 0.0})), ("@united it was UA flight 1001! Now I'm currently stuck in
Portland because I missed my ride because of the 3 hour delay. #GetMartyHome",
{'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0})), ('@SouthwestAir nice
touch on the passbook integration!', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0,
'compound': 0.0})), ('Keep it up :) @AmericanAir', {'neg': 0.0, 'neu': 0.0,
'pos': 0.0, 'compound': 0.0})), ('@AmericanAir DONE!! thank you so much!!',
{'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0})), ('@USAirways any chance
of a travel advisory or DCA this weekend?', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0,
'compound': 0.0})), ('@JetBlue keep going back and forth with being able to board
and then not. But great agents in savannah!', {'neg': 0.0, 'neu': 0.0, 'pos':
0.0, 'compound': 0.0})), ('"@JetBlue Flight 152 (N559JB) "'Here\'s Looking at
Blue Kid"' departs @MCO enroute to @BostonLogan http://t.co/qeaDA92MW6"',
{'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0})), ('"@united diverged to
Burlington', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0})), ('@united
thank you for taking care of my mom and reFlight Booking Problems her flight in
phl.', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0})), ("@united thanks
we'll be in touch!", {'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0})),
("@united line full of worried customers and guy at desk says so sorry you will
have to take care of your problems Late Flightr I don't have time.", {'neg':
0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0}]]
```

[167]: *# Running VADER searching for the adjectives on lemmatized text*

```
lemmatized_adjective_text_lst = []

for sentence in airline_tweets_train.data:
    lemmatized_adjective_text = sentence.decode("utf-8")
    lemmatized_adjective_output = run_vader(lemmatized_adjective_text,
    ↪lemmatize=True, parts_of_speech_to_consider={'JJ'}, verbose=0)
    lemmatized_adjective_text_lst.append((lemmatized_adjective_text,
    ↪lemmatized_adjective_output))
print(lemmatized_adjective_text_lst[0:30])
```

```
[('@AmericanAir Why is your cover photo of TWA? Just wondering.', {'neg': 0.0,
'neu': 0.0, 'pos': 0.0, 'compound': 0.0})), ('@united next day', {'neg': 0.0,
'neu': 0.0, 'pos': 0.0, 'compound': 0.0})), ('"@JetBlue Thank you that it is not
just a livery; it is a culture that 16', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0,
'compound': 0.0})), ('@SouthwestAir I cannot DM you as you do not follow me.',
{'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0})), ('"@VirginAmerica Funny
story', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0})), ('@united flight
403 is delayed 40 min bc a missing screw and were in danger of missing flight
6491', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0})), ('@AmericanAir
does it show that we are on the same plane?', {'neg': 0.0, 'neu': 0.0, 'pos':
```

```

0.0, 'compound': 0.0})), ("@united @perfectomobile You really shouldn't use the
word 'crashing.' Just sayin'."), {'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound':
0.0})), ("@united what's a good number to call to speak with someone about how
you can fix what you did to 50 people and their luggage on Saturday?", {'neg':
0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0})), ('@USAirways Im researching a
flight & noticed a small issue w/class of service between your site &
other sites. Could Reservations help me?', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0,
'compound': 0.0})), ('"@united never fails', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0,
'compound': 0.0})), ('"@united hi guys', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0,
'compound': 0.0})), ('"@united broken light', {'neg': 0.0, 'neu': 0.0, 'pos':
0.0, 'compound': 0.0})), ('@united to speak to a real person to get this resolved
politely and efficiently. (2/2)', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0,
'compound': 0.0})), ('@united love the new 1st class breakfast!', {'neg': 0.0,
'neu': 0.0, 'pos': 0.0, 'compound': 0.0})), ('"@USAirways just heard #Monterey
airport was closed due to thick fog', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0,
'compound': 0.0})), ('@united I did DM the details but still no response. I have
to book today.', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0})),
("@united yes 1427 Cancelled Flightled. Moved to 333. I'll figure out car rental
changes and such.", {'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0})),
('"@USAirways awesome! And yes', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0,
'compound': 0.0})), ("@united it was UA flight 1001! Now I'm currently stuck in
Portland because I missed my ride because of the 3 hour delay. #GetMartyHome",
{'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0})), ('@SouthwestAir nice
touch on the passbook integration!', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0,
'compound': 0.0})), ('Keep it up :) @AmericanAir', {'neg': 0.0, 'neu': 0.0,
'pos': 0.0, 'compound': 0.0})), ('@AmericanAir DONE!! thank you so much!!',
{'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0})), ('@USAirways any chance
of a travel advisory or DCA this weekend?', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0,
'compound': 0.0})), ('@JetBlue keep going back and forth with being able to board
and then not. But great agents in savannah!', {'neg': 0.0, 'neu': 0.0, 'pos':
0.0, 'compound': 0.0})), ('"@JetBlue Flight 152 (N559JB) ""Here\'s Looking at
Blue Kid"" departs @MCO enroute to @BostonLogan http://t.co/qeaDA92MW6"',
{'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0})), ('"@united diverged to
Burlington', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0})), ('@united
thank you for taking care of my mom and reFlight Booking Problems her flight in
phl.', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0})), ("@united thanks
we'll be in touch!", {'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0})),
("@united line full of worried customers and guy at desk says so sorry you will
have to take care of your problems Late Flightr I don't have time.", {'neg':
0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0}]]

```

[165]: *# Running VADER searching for nouns*

```

noun_text_lst = []

for sentence in airline_tweets_train.data:
    noun_text = sentence.decode("utf-8")

```



```

noun_output = run_vader(noun_text, lemmatize=False,
↳parts_of_speech_to_consider={'NOUN'}, verbose=0)
noun_text_lst.append((noun_text, noun_output))
print(noun_text_lst[0:30])

```

```

[('@AmericanAir Why is your cover photo of TWA? Just wondering.', {'neg': 0.0,
'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('@united next day', {'neg': 0.0,
'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('"@JetBlue Thank you that it is not
just a livery; it is a culture that 16', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0,
'compound': 0.0}), ('@SouthwestAir I cannot DM you as you do not follow me.',
{'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('"@VirginAmerica Funny
story', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('@united flight
403 is delayed 40 min bc a missing screw and were in danger of missing flight
6491', {'neg': 0.615, 'neu': 0.385, 'pos': 0.0, 'compound': -0.5859}),
('@AmericanAir does it show that we are on the same plane?', {'neg': 0.0, 'neu':
1.0, 'pos': 0.0, 'compound': 0.0}), ('@united @perfectomobile You really
shouldn't use the word 'crashing.' Just sayin'.', {'neg': 0.0, 'neu': 1.0,
'pos': 0.0, 'compound': 0.0}), ('@united what's a good number to call to speak
with someone about how you can fix what you did to 50 people and their luggage
on Saturday?', {'neg': 0.0, 'neu': 0.606, 'pos': 0.394, 'compound': 0.0772}),
('@USAirways Im researching a flight & noticed a small issue w/class of
service between your site & other sites. Could Reservations help me?',
{'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('"@united never fails',
{'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0}), ('"@united hi guys',
{'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('"@united broken
light', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('@united to
speak to a real person to get this resolved politely and efficiently. (2/2)',
{'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('@united love the new
1st class breakfast!', {'neg': 0.0, 'neu': 0.323, 'pos': 0.677, 'compound':
0.6369}), ('"@USAirways just heard #Monterey airport was closed due to thick
fog', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('@united I did DM
the details but still no response. I have to book today.', {'neg': 0.0, 'neu':
1.0, 'pos': 0.0, 'compound': 0.0}), ('@united yes 1427 Cancelled Flightled.
Moved to 333. I'll figure out car rental changes and such.', {'neg': 0.0, 'neu':
1.0, 'pos': 0.0, 'compound': 0.0}), ('"@USAirways awesome! And yes', {'neg':
0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0}), ('@united it was UA flight 1001!
Now I'm currently stuck in Portland because I missed my ride because of the 3
hour delay. #GetMartyHome", {'neg': 0.365, 'neu': 0.635, 'pos': 0.0, 'compound':
-0.3182}), ('@SouthwestAir nice touch on the passbook integration!', {'neg':
0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('Keep it up :) @AmericanAir',
{'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0}), ('@AmericanAir DONE!!
thank you so much!!', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0}),
('@USAirways any chance of a travel advisory or DCA this weekend?', {'neg': 0.0,
'neu': 0.667, 'pos': 0.333, 'compound': 0.25}), ('@JetBlue keep going back and
forth with being able to board and then not. But great agents in savannah!',
{'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('"@JetBlue Flight 152
(N559JB) ""Here\'s Looking at Blue Kid"" departs @MCO enroute to @BostonLogan
http://t.co/qeaDA92MW6"', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound':

```

```
0.0})), ('"@united diverged to Burlington', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0,
'compound': 0.0})), ('@united thank you for taking care of my mom and reFlight
Booking Problems her flight in phl.', {'neg': 0.248, 'neu': 0.459, 'pos': 0.294,
'compound': 0.128})), ('"@united thanks we'll be in touch!", {'neg': 0.0, 'neu':
0.256, 'pos': 0.744, 'compound': 0.4404})), ('"@united line full of worried
customers and guy at desk says so sorry you will have to take care of your
problems Late Flightr I don't have time.", {'neg': 0.248, 'neu': 0.459, 'pos':
0.294, 'compound': 0.128}]]
```

[164]: *# Running VADER searching for nouns on lemmatized text*

```
lemma_noun_text_lst = []

for sentence in airline_tweets_train.data:
    lemma_noun_text = sentence.decode("utf-8")
    lemma_noun_output = run_vader(lemma_noun_text, lemmatize=True,
↳parts_of_speech_to_consider={'NOUN'}, verbose=0)
    lemma_noun_text_lst.append((lemma_noun_text, lemma_noun_output))
print(lemma_noun_text_lst[0:30])
```

```
[('@AmericanAir Why is your cover photo of TWA? Just wondering.', {'neg': 0.0,
'neu': 1.0, 'pos': 0.0, 'compound': 0.0})), ('@united next day', {'neg': 0.0,
'neu': 1.0, 'pos': 0.0, 'compound': 0.0})), ('"@JetBlue Thank you that it is not
just a livery; it is a culture that 16', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0,
'compound': 0.0})), ('@SouthwestAir I cannot DM you as you do not follow me.',
{'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0})), ('"@VirginAmerica Funny
story', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0})), ('@united flight
403 is delayed 40 min bc a missing screw and were in danger of missing flight
6491', {'neg': 0.615, 'neu': 0.385, 'pos': 0.0, 'compound': -0.5859})),
('@AmericanAir does it show that we are on the same plane?', {'neg': 0.0, 'neu':
1.0, 'pos': 0.0, 'compound': 0.0})), ('@united @perfectomobile You really
shouldn't use the word 'crashing.' Just sayin'.", {'neg': 0.0, 'neu': 1.0,
'pos': 0.0, 'compound': 0.0})), ('"@united what's a good number to call to speak
with someone about how you can fix what you did to 50 people and their luggage
on Saturday?', {'neg': 0.0, 'neu': 0.606, 'pos': 0.394, 'compound': 0.0772})),
('@USAirways Im researching a flight & noticed a small issue w/class of
service between your site & other sites. Could Reservations help me?',
{'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0})), ('"@united never fails',
{'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0})), ('"@united hi guys',
{'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0})), ('"@united broken
light', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0})), ('@united to
speak to a real person to get this resolved politely and efficiently. (2/2)',
{'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0})), ('@united love the new
1st class breakfast!', {'neg': 0.0, 'neu': 0.323, 'pos': 0.677, 'compound':
0.6369})), ('"@USAirways just heard #Monterey airport was closed due to thick
fog', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0})), ('@united I did DM
the details but still no response. I have to book today.', {'neg': 0.0, 'neu':
1.0, 'pos': 0.0, 'compound': 0.0})), ('@united yes 1427 Cancelled Flightled.
```

```
Moved to 333. I'll figure out car rental changes and such.", {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('"@USAirways awesome! And yes', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0}), ("@united it was UA flight 1001! Now I'm currently stuck in Portland because I missed my ride because of the 3 hour delay. #GetMartyHome", {'neg': 0.365, 'neu': 0.635, 'pos': 0.0, 'compound': -0.3182}), ('@SouthwestAir nice touch on the passbook integration!', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('Keep it up :) @AmericanAir', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0}), ('@AmericanAir DONE!! thank you so much!!', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0}), ('@USAirways any chance of a travel advisory or DCA this weekend?', {'neg': 0.0, 'neu': 0.667, 'pos': 0.333, 'compound': 0.25}), ('@JetBlue keep going back and forth with being able to board and then not. But great agents in savannah!', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('"@JetBlue Flight 152 (N559JB) ""Here\'s Looking at Blue Kid"" departs @MCO enroute to @BostonLogan http://t.co/qeaDA92MW6"', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('"@united diverged to Burlington', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('@united thank you for taking care of my mom and reFlight Booking Problems her flight in phl.', {'neg': 0.248, 'neu': 0.459, 'pos': 0.294, 'compound': 0.128}), ("@united thanks we'll be in touch!", {'neg': 0.0, 'neu': 0.286, 'pos': 0.714, 'compound': 0.3612}), ("@united line full of worried customers and guy at desk says so sorry you will have to take care of your problems Late Flightr I don't have time.", {'neg': 0.248, 'neu': 0.459, 'pos': 0.294, 'compound': 0.128})]
```

[169]: *# Running VADER searching for verbs*

```
verb_text_lst = []

for sentence in airline_tweets_train.data:
    verb_text = sentence.decode("utf-8")
    verb_output = run_vader(verb_text, lemmatize=False,
↪parts_of_speech_to_consider={'NOUN'}, verbose=0)
    verb_text_lst.append((verb_text, verb_output))
print(verb_text_lst[0:30])
```

```
[('@AmericanAir Why is your cover photo of TWA? Just wondering.', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('@united next day', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('"@JetBlue Thank you that it is not just a livery; it is a culture that 16', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('@SouthwestAir I cannot DM you as you do not follow me.', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('"@VirginAmerica Funny story', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('@united flight 403 is delayed 40 min bc a missing screw and were in danger of missing flight 6491', {'neg': 0.615, 'neu': 0.385, 'pos': 0.0, 'compound': -0.5859}), ('@AmericanAir does it show that we are on the same plane?', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ("@united @perfectomobile You really shouldn't use the word 'crashing.' Just sayin'." , {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ("@united what's a good number to call to speak
```


with someone about how you can fix what you did to 50 people and their luggage on Saturday?", {'neg': 0.0, 'neu': 0.606, 'pos': 0.394, 'compound': 0.0772}), ('@USAirways Im researching a flight & noticed a small issue w/class of service between your site & other sites. Could Reservations help me?', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('"@united never fails', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0}), ('"@united hi guys', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('"@united broken light', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('@united to speak to a real person to get this resolved politely and efficiently. (2/2)', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('@united love the new 1st class breakfast!', {'neg': 0.0, 'neu': 0.323, 'pos': 0.677, 'compound': 0.6369}), ('"@USAirways just heard #Monterey airport was closed due to thick fog', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('@united I did DM the details but still no response. I have to book today.', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('"@united yes 1427 Cancelled Flightled. Moved to 333. I'll figure out car rental changes and such.", {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('"@USAirways awesome! And yes', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0}), ('"@united it was UA flight 1001! Now I'm currently stuck in Portland because I missed my ride because of the 3 hour delay. #GetMartyHome", {'neg': 0.365, 'neu': 0.635, 'pos': 0.0, 'compound': -0.3182}), ('@SouthwestAir nice touch on the passbook integration!', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('Keep it up :) @AmericanAir', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0}), ('@AmericanAir DONE!! thank you so much!!', {'neg': 0.0, 'neu': 0.0, 'pos': 0.0, 'compound': 0.0}), ('@USAirways any chance of a travel advisory or DCA this weekend?', {'neg': 0.0, 'neu': 0.667, 'pos': 0.333, 'compound': 0.25}), ('@JetBlue keep going back and forth with being able to board and then not. But great agents in savannah!', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('"@JetBlue Flight 152 (N559JB) ""Here\'s Looking at Blue Kid"" departs @MCO enroute to @BostonLogan <http://t.co/qeaDA92MW6>", {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('"@united diverged to Burlington', {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}), ('@united thank you for taking care of my mom and reFlight Booking Problems her flight in phl.', {'neg': 0.248, 'neu': 0.459, 'pos': 0.294, 'compound': 0.128}), ('"@united thanks we'll be in touch!", {'neg': 0.0, 'neu': 0.256, 'pos': 0.744, 'compound': 0.4404}), ('"@united line full of worried customers and guy at desk says so sorry you will have to take care of your problems Late Flightr I don't have time.", {'neg': 0.248, 'neu': 0.459, 'pos': 0.294, 'compound': 0.128})]

3.3 Part II: scikit-learn assignments

3.3.1 [4 points] Question 5

Train the scikit-learn classifier (Naive Bayes) using the airline tweets.

- Train the model on the airline tweets with 80% training and 20% test set and default settings (TF-IDF representation, min_df=2)
- Train with different settings:
 - with respect to vectorizing: TF-IDF ('airline_tfidf') vs. Bag of words representation

- (‘airline_count’)
 - with respect to the frequency threshold (min_df). Carry out experiments with increasing values for document frequency (min_df = 2; min_df = 5; min_df = 10)
- [1 point] a. Generate a classification_report for all experiments
- [3 points] b. Look at the results of the experiments with the different settings and try to explain why they differ:
 - which category performs best, is this the case for any setting?
 - does the frequency threshold affect the scores? Why or why not according to you?

```
[131]: from sklearn.feature_extraction.text import CountVectorizer, TfidfVectorizer
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB
from sklearn.datasets import load_files

# Load airline tweets dataset
airline_tweets = load_files(path, encoding='utf-8', decode_error='ignore')

# Split data into training (80%) and testing (20%)
X_train, X_test, y_train, y_test = train_test_split(
    airline_tweets.data, airline_tweets.target, test_size=0.2, random_state=42)

# Experiment configurations
vectorizer_options = {'airline_tfidf': TfidfVectorizer, 'airline_count_bof':
    ↪CountVectorizer}
min_df_values = [2, 5, 10]

# Run experiments
results = {}
for vec_name, VecClass in vectorizer_options.items():
    for min_df in min_df_values:
        vectorizer = VecClass(min_df=min_df)
        X_train_vec = vectorizer.fit_transform(X_train)
        X_test_vec = vectorizer.transform(X_test)

        model = MultinomialNB()
        model.fit(X_train_vec, y_train)
        y_pred = model.predict(X_test_vec)

        report = classification_report(y_test, y_pred,
    ↪target_names=airline_tweets.target_names)
        results[(vec_name, min_df)] = report

# Print results
for (vec_name, min_df), report in results.items():
    print(f"\n=== Results for {vec_name} with min_df={min_df} ===")
    print(report)
```

```

=== Results for airline_tfidf with min_df=2 ===
      precision    recall  f1-score   support

negative      0.82      0.94      0.88      335
neutral       0.85      0.74      0.79      305
positive      0.85      0.83      0.84      311

accuracy                    0.84      951
macro avg      0.84      0.84      0.84      951
weighted avg   0.84      0.84      0.84      951

```

```

=== Results for airline_tfidf with min_df=5 ===
      precision    recall  f1-score   support

negative      0.82      0.93      0.87      335
neutral       0.82      0.74      0.78      305
positive      0.83      0.80      0.82      311

accuracy                    0.83      951
macro avg      0.83      0.82      0.82      951
weighted avg   0.83      0.83      0.82      951

```

```

=== Results for airline_tfidf with min_df=10 ===
      precision    recall  f1-score   support

negative      0.82      0.91      0.86      335
neutral       0.80      0.75      0.78      305
positive      0.84      0.79      0.82      311

accuracy                    0.82      951
macro avg      0.82      0.82      0.82      951
weighted avg   0.82      0.82      0.82      951

```

```

=== Results for airline_count_bof with min_df=2 ===
      precision    recall  f1-score   support

negative      0.84      0.93      0.89      335
neutral       0.86      0.75      0.80      305
positive      0.85      0.85      0.85      311

accuracy                    0.85      951
macro avg      0.85      0.84      0.84      951
weighted avg   0.85      0.85      0.85      951

```

```

=== Results for airline_count_bof with min_df=5 ===
      precision    recall  f1-score   support

negative      0.84      0.93      0.88      335
neutral       0.83      0.76      0.80      305
positive      0.85      0.82      0.83      311

accuracy                    0.84      951
macro avg      0.84      0.84      0.84      951
weighted avg   0.84      0.84      0.84      951

```

```

=== Results for airline_count_bof with min_df=10 ===
      precision    recall  f1-score   support

negative      0.83      0.92      0.87      335
neutral       0.82      0.76      0.79      305
positive      0.84      0.80      0.82      311

accuracy                    0.83      951
macro avg      0.83      0.82      0.82      951
weighted avg   0.83      0.83      0.83      951

```

4 Analysis of results

b)i) Looking at precision, the categories perform similarly with positive performing slightly better with exception of $df = 2$ where neutral performs slightly higher. Recall scores on the other hand are more distinct with negative performing the best, positive performing slightly worse and neutral performing the worst. This case remains the same for most of the settings. ii) Different settings created differing scores, however the scores didn't differ significantly (like ± 0.02) so a conclusion can be drawn that different frequency settings significantly affect the scores. *

4.0.1 [4 points] Question 6: Inspecting the best scoring features

- Train the scikit-learn classifier (Naive Bayes) model with the following settings (airline tweets 80% training and 20% test; Bag of words representation ('airline_count'), min_df=2)
- [1 point] a. Generate the list of best scoring features per class (see function **important_features_per_class** below) [1 point]
- [3 points] b. Look at the lists and consider the following issues:
 - [1 point] Which features did you expect for each separate class and why?
 - [1 point] Which features did you not expect and why ?
 - [1 point] The list contains all kinds of words such as names of airlines, punctuation, numbers and content words (e.g., 'delay' and 'bad'). Which words would you remove or keep when trying to improve the model and why?

```

[132]: import numpy as np

def important_features_per_class(vectorizer, classifier, n=80):
    class_labels = classifier.classes_
    feature_names = vectorizer.get_feature_names_out()
    topn_class1 = sorted(zip(classifier.feature_count_[0],
    ↪ feature_names), reverse=True)[:n]
    topn_class2 = sorted(zip(classifier.feature_count_[1],
    ↪ feature_names), reverse=True)[:n]
    topn_class3 = sorted(zip(classifier.feature_count_[2],
    ↪ feature_names), reverse=True)[:n]
    print("Important words in negative documents")
    for coef, feat in topn_class1:
        print(class_labels[0], coef, feat)
    print("-----")
    print("Important words in neutral documents")
    for coef, feat in topn_class2:
        print(class_labels[1], coef, feat)
    print("-----")
    print("Important words in positive documents")
    for coef, feat in topn_class3:
        print(class_labels[2], coef, feat)

# example of how to call from notebook:
important_features_per_class(airline_vec, clf)

# Split data into training (80%) and testing (20%)
X_train, X_test, y_train, y_test = train_test_split(
    airline_tweets.data, airline_tweets.target, test_size=0.2, random_state=42)

# Bag of Words Representation with min_df=2
vectorizer = CountVectorizer(min_df=2)
X_train_vec = vectorizer.fit_transform(X_train)
X_test_vec = vectorizer.transform(X_test)

# Train Naive Bayes Model
model = MultinomialNB()
model.fit(X_train_vec, y_train)

# Ensure feature_count_ is correctly assigned
model.feature_count_ = np.exp(model.feature_log_prob_) # Convert log_
    ↪ probabilities to actual feature counts

# Generate classification report
y_pred = model.predict(X_test_vec)
report = classification_report(y_test, y_pred, target_names=airline_tweets.
    ↪ target_names)

```

```
print(report)

# Call function to print important features
important_features_per_class(vectorizer, model)
```

	precision	recall	f1-score	support
negative	0.84	0.93	0.89	335
neutral	0.86	0.75	0.80	305
positive	0.85	0.85	0.85	311
accuracy			0.85	951
macro avg	0.85	0.84	0.84	951
weighted avg	0.85	0.85	0.85	951

Important words in negative documents

```
0 0.05842987996152074 united
0 0.031954494123551794 to
0 0.023003889748630203 the
0 0.016939227905809512 flight
0 0.015517169266803287 you
0 0.015475344012714866 and
0 0.013844159103266546 on
0 0.01334225605420552 for
0 0.01296582876740976 my
0 0.012505750972437158 is
0 0.012003847923376132 in
0 0.009076080137186832 your
0 0.00895060437492157 of
0 0.008406876071772132 it
0 0.007779497260445857 that
0 0.007152118449119578 not
0 0.006901166924589062 me
0 0.006650215400058552 have
0 0.006608390145970134 with
0 0.006441089129616462 no
0 0.0061483123509975274 was
0 0.005981011334643853 at
0 0.0058137103182901795 this
0 0.005144506252875482 service
0 0.004935379982433393 can
0 0.004475302187460785 virginamerica
0 0.004475302187460785 be
0 0.004433476933372367 get
0 0.004391651679283948 from
0 0.00434982642519553 we
0 0.00434982642519553 now
0 0.004057049646576601 cancelled
```

0 0.004015224392488182 customer
0 0.003931573884311345 an
0 0.0037224476138692517 they
0 0.0037224476138692517 delayed
0 0.0037224476138692517 been
0 0.0036387971056924154 bag
0 0.0035969718516039952 plane
0 0.003513321343427159 why
0 0.0034714960893387423 but
0 0.003387845581161905 just
0 0.003387845581161905 are
0 0.003304195072985067 time
0 0.003262369818896648 co
0 0.003178719310719811 so
0 0.0030950688025429755 http
0 0.003053243548454555 hours
0 0.002885942532100882 what
0 0.002802292023924043 gate
0 0.0027604667698356245 when
0 0.002718641515747206 airline
0 0.0026349910075703694 still
0 0.0026349910075703694 has
0 0.0026349910075703694 amp
0 0.0025931657534819515 will
0 0.0025931657534819515 hour
0 0.002551340499393533 up
0 0.0025095152453051134 late
0 0.0025095152453051134 again
0 0.0024676899912166963 out
0 0.0024676899912166963 help
0 0.002425864737128276 how
0 0.002425864737128276 do
0 0.002425864737128276 about
0 0.0023422142289514407 worst
0 0.0023422142289514407 our
0 0.0023422142289514407 flights
0 0.002300388974863022 after
0 0.002258563720774603 had
0 0.0022167384666861856 waiting
0 0.002174913212597765 would
0 0.002133087958509348 or
0 0.002133087958509348 like
0 0.002133087958509348 delay
0 0.002091262704420929 one
0 0.002091262704420929 never
0 0.002091262704420929 as
0 0.0020494374503325103 if
0 0.0020494374503325103 because

Important words in neutral documents

1 0.038322211630123955 to
1 0.019256434699714015 jetblue
1 0.018112488083889433 the
1 0.01703209405783286 united
1 0.01665077851922466 southwestair
1 0.015697489672704176 on
1 0.01537972672386401 flight
1 0.015061963775023845 you
1 0.012837623133142686 my
1 0.012138544645694319 for
1 0.012011439466158255 co
1 0.011566571337782019 americanair
1 0.011248808388941852 http
1 0.011121703209405793 in
1 0.010549729901493495 is
1 0.010486177311725456 usairways
1 0.010359072132189392 can
1 0.00946933587543693 and
1 0.008706704798220534 from
1 0.0080711789005402 it
1 0.007626310772163968 of
1 0.0068001271051795365 have
1 0.006355258976803311 me
1 0.005719733079122977 do
1 0.005147759771210681 with
1 0.005020654591674613 get
1 0.00495710200190658 virginamerica
1 0.00495710200190658 any
1 0.004893549412138549 that
1 0.004639339053066414 this
1 0.0045757864632983815 what
1 0.004512233873530349 at
1 0.004448681283762317 if
1 0.004448681283762317 be
1 0.004194470924690182 are
1 0.004130918334922149 we
1 0.004067365745154115 flights
1 0.003940260565618052 will
1 0.003876707975850018 help
1 0.0037496027963139526 there
1 0.003558945027009854 need
1 0.0034953924372418204 please
1 0.003431839847473787 our
1 0.003304734667937719 your
1 0.003241182078169688 just
1 0.003177629488401655 when

1 0.003114076898633622 how
1 0.003050524308865587 us
1 0.0026056561804893566 was
1 0.0026056561804893566 out
1 0.0026056561804893566 not
1 0.0026056561804893566 dm
1 0.002542103590721322 tomorrow
1 0.00247855100095329 know
1 0.00247855100095329 an
1 0.0024149984111852566 so
1 0.0024149984111852566 now
1 0.0023514458214172226 thanks
1 0.0022243406418811563 or
1 0.0022243406418811563 hi
1 0.0022243406418811563 fleet
1 0.0022243406418811563 fleek
1 0.0022243406418811563 am
1 0.0022243406418811563 about
1 0.002160788052113125 would
1 0.002160788052113125 flying
1 0.002097235462345091 fly
1 0.002097235462345091 change
1 0.0020336828725770594 way
1 0.0019701302828090254 like
1 0.0019701302828090254 cancelled
1 0.0019701302828090254 amp
1 0.0018430251032729604 one
1 0.0018430251032729604 as
1 0.0017794725135049268 by
1 0.0017794725135049268 but
1 0.0017159199237368932 up
1 0.0017159199237368932 they
1 0.0017159199237368932 number
1 0.0017159199237368932 no

Important words in positive documents

2 0.03003452993680371 you
2 0.027233044498012907 the
2 0.026907290377223288 to
2 0.021630073620431296 for
2 0.019349794774903912 southwestair
2 0.018893739005798424 jetblue
2 0.018111929115903317 thanks
2 0.016222555215323474 united
2 0.01609225356700763 thank
2 0.013160466479900977 and
2 0.012313505765847943 flight
2 0.010684735161899803 americanair

2 0.009968076096162617 my
2 0.009642321975372988 on
2 0.00931656785458336 usairways
2 0.008795361261319959 great
2 0.008209003843898628 in
2 0.007948400547266926 your
2 0.007687797250635225 it
2 0.007036289009055968 is
2 0.006840836536582187 of
2 0.006840836536582187 me
2 0.006254479119160858 with
2 0.006189328295002933 was
2 0.006189328295002933 so
2 0.005928724998371229 at
2 0.005733272525897456 virginamerica
2 0.005668121701739528 service
2 0.005277216756791977 co
2 0.005016613460160273 love
2 0.00488631181184442 this
2 0.004821160987686497 http
2 0.004495406866896869 just
2 0.004495406866896869 are
2 0.004234803570265164 from
2 0.004104501921949313 customer
2 0.004039351097791389 best
2 0.004039351097791389 awesome
2 0.003974200273633463 guys
2 0.003909049449475535 much
2 0.0037787478011596844 that
2 0.0034529936803700595 have
2 0.0034529936803700595 all
2 0.003387842856212131 time
2 0.0033226920320542077 good
2 0.0033226920320542077 be
2 0.003257541207896282 we
2 0.003192390383738356 airline
2 0.0030620887354225034 up
2 0.0030620887354225034 amazing
2 0.0029969379112645784 got
2 0.002736334614632875 out
2 0.0026711837904749504 today
2 0.0026711837904749504 help
2 0.0026711837904749504 amp
2 0.0025408821421590987 our
2 0.0025408821421590987 get
2 0.0025408821421590987 crew
2 0.002475731318001173 us
2 0.002475731318001173 now

2 0.0024105804938432465 very
 2 0.002345429669685322 will
 2 0.0022802788455273952 can
 2 0.0022802788455273952 but
 2 0.0021499771972115444 they
 2 0.0021499771972115444 made
 2 0.0021499771972115444 fly
 2 0.0020848263730536207 do
 2 0.0020848263730536207 an
 2 0.002019675548895694 work
 2 0.002019675548895694 not
 2 0.002019675548895694 gate
 2 0.002019675548895694 flying
 2 0.002019675548895694 appreciate
 2 0.0018242230764219177 home
 2 0.0017590722522639916 see
 2 0.0017590722522639916 flights
 2 0.0017590722522639916 back
 2 0.0016939214281060653 she
 2 0.0016939214281060653 response

Expected features:

For the negative class the words: cancelled, delayed, no, and never; are expected because these words have negative connotations even when used by themselves.

For the neutral class the words: from, have, with, that, this; are expected because these words do not have any connotations by themselves and therefore are usually used as facilitator words.

For the positive class the words: thank, great, love, best, awesome, good, amazing, appreciate; are expected because these words by themselves have a positive connotation when a human thinks about them.

Unexpected features:

For the negative class the words: now, customer, just, what, because; are not expected, these words are very neutral and do not have a negative connotation associated with them, therefore, it is interesting that they became classified in this category.

For the neutral class the words: not, no; are also not expected, although these can be interpreted as neutral words, they are usually negations and negations are mostly used in negative circumstances instead of neutral or positive circumstances.

For the positive class the words: service, this, today, not, response; are not expected, these words are very neutral in their nature and do not have any sort of indicative connotation, therefore, it is also interesting that they are classed in this category.

When trying to improve the model, removing such words as: response, today, service, just, because, etc. would help the model since avoiding having the same word be useful in many categories allowed the model to make better distinctions; like this, the model is not as deterministic when classifying tweets.

4.0.2 [Optional! (will not be graded)] Question 7

Train the model on airline tweets and test it on your own set of tweets + Train the model with the following settings (airline tweets 80% training and 20% test; Bag of words representation ('airline_count'), min_df=2) + Apply the model on your own set of tweets and generate the classification report * [1 point] a. Carry out a quantitative analysis. * [1 point] b. Carry out an error analysis on 10 correctly and 10 incorrectly classified tweets and discuss them * [2 points] c. Compare the results (cf. classification report) with the results obtained by VADER on the same tweets and discuss the differences.

```
[116]: # Split data into training (80%) and testing (20%)
X_train, X_test, y_train, y_test = train_test_split(
    airline_tweets_train.data, airline_tweets_train.target, test_size=0.2,
    random_state=42)

# Bag of Words Representation with min_df=2
vectorizer = CountVectorizer(min_df=2)
X_train_vec = vectorizer.fit_transform(X_train)
X_test_vec = vectorizer.transform(X_test)

# Train Naive Bayes Model
model = MultinomialNB()
model.fit(X_train_vec, y_train)

# Evaluate on test set
y_pred = model.predict(X_test_vec)
print("Classification Report on Airline Tweets:")
print(classification_report(y_test, y_pred, target_names=airline_tweets_train.
    target_names))

tweet_texts = [tweet["text_of_tweet"] for tweet in my_tweets.values()]
X_my_tweets_vec = vectorizer.transform(tweet_texts)

# Predict sentiment for user tweets
y_my_pred = model.predict(X_my_tweets_vec)
print("Classification Report on My Tweets:")
print(classification_report([0] * len(y_my_pred), y_my_pred,
    target_names=airline_tweets_train.target_names)) # Assuming neutral baseline

# VADER Sentiment Analysis
sia = SentimentIntensityAnalyzer()
vader_results = ["pos" if sia.polarity_scores(text)["compound"] > 0.05 else
    "neg" if sia.polarity_scores(text)["compound"] < -0.05 else "neu" for text
    in tweet_texts]

# Compare VADER with Naive Bayes
print("Comparison with VADER:")
```

```
print(f"VADER Sentiments: {vader_results[:10]}")
print(f"Naive Bayes Predictions: {y_my_pred[:10]}")
```

Classification Report on Airline Tweets:

	precision	recall	f1-score	support
negative	0.84	0.93	0.89	335
neutral	0.86	0.75	0.80	305
positive	0.85	0.85	0.85	311
accuracy			0.85	951
macro avg	0.85	0.84	0.84	951
weighted avg	0.85	0.85	0.85	951

Classification Report on My Tweets:

	precision	recall	f1-score	support
negative	1.00	0.60	0.75	50
neutral	0.00	0.00	0.00	0
positive	0.00	0.00	0.00	0
accuracy			0.60	50
macro avg	0.33	0.20	0.25	50
weighted avg	1.00	0.60	0.75	50

Comparison with VADER:

VADER Sentiments: ['pos', 'neg', 'neg', 'neg', 'pos', 'neg', 'neg', 'neg', 'pos', 'neg']

Naive Bayes Predictions: [0 0 0 1 0 0 2 0 2 1]

/Users/robertostoica/Desktop/anaconda3/envs/Text_Mining_python/lib/python3.12/site-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Recall is ill-defined and being set to 0.0 in labels with no true samples. Use `zero\_division` parameter to control this behavior.

_warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))

/Users/robertostoica/Desktop/anaconda3/envs/Text_Mining_python/lib/python3.12/site-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Recall is ill-defined and being set to 0.0 in labels with no true samples. Use `zero\_division` parameter to control this behavior.

_warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))

/Users/robertostoica/Desktop/anaconda3/envs/Text_Mining_python/lib/python3.12/site-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Recall is ill-defined and being set to 0.0 in labels with no true samples. Use `zero\_division` parameter to control this behavior.

_warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))

```
[133]: incorrect_positive = []
        incorrect_negative = []
```

```

incorrect_neutral = []

for tweet, pred, actual in zip(tweet_texts, vader_results, gold):
    if pred != actual:
        if actual == "positive":
            incorrect_positive.append(tweet)
        elif actual == "negative":
            incorrect_negative.append(tweet)
        elif actual == "neutral":
            incorrect_neutral.append(tweet)

# Show misclassified tweets (up to 10 per category)
print("Incorrectly classified positive tweets:", incorrect_positive[:10],
      ↪len(incorrect_positive), 'tweets')
print()
print("Incorrectly classified negative tweets:", incorrect_negative[:10],
      ↪len(incorrect_negative), 'tweets')
print()
print("Incorrectly classified neutral tweets:", incorrect_neutral[:10],
      ↪len(incorrect_neutral), 'tweets')

```

Incorrectly classified positive tweets: ['The Black Panther closed his eyes & just seemed so content. My heart melted', 'Rocket, the majestic male tiger of the Pilibhit Tiger Reserve, in India truly lives up to his bold and regal appearance', 'This is so beautiful it's almost scary', 'A beautiful bond between K-9 and his handler', 'I'll say, it's nice to have someone actually say this', 'Thirty-two years together and you still take my breath away. Happy Valentine's Day', '10 terrifying beauties of wildlife', 'Gorgeous grey wolf becomes a good boy when visited by the people that helped raise their pack', 'THIS DOG AND DEER BEING FRIENDS IS LITERALLY THE CUTEST THING EVER ', 'I have no idea what is happening here but it is the best thing I've seen today'] 15 tweets

Incorrectly classified negative tweets: ['michelle trachtenberg has sadly passed away at the age of 39. rest in peace', 'Your a War Criminal - Thousands are dead because of you', 'I guess this is a controversial opinion but men can be raped and abused by women', 'If Elon didn't buy Twitter, it's entirely possible that Kamala would be president right now. Terrifying thought', 'Leaking ICE raid details is not just betrayal, it is a national security risk', 'Nazis and racists get to spew hate all day on this app but NBA Centel does a little trolling and gets banned. interesting', 'Dad looking annoyed and mum crying with disappointment that it's a girl', 'As a father this has to be your worst nightmare', 'Complete batshit lunacy on the president's Truth Social account', 'Nigga that got killed by that white lady digital footprint is HORRIBLE'] 18 tweets

Incorrectly classified neutral tweets: ['I'm convinced that pandas are just drunk people in costumes', 'Animals reacting to themselves in the mirror in

jungle', 'Penguin paints a picture and is extremely pleased with herself', 'The professor banned laptops so the students had to find a way', 'Unfortunately, as President Bukele eloquently articulates, there is no other option', 'On this day one year ago, TikToker mr.weatherall posted the following video', 'rest in peace michelle trachtenberg. thank you for the childhood memories', 'I regret to inform you that this site still has the juice', 'This is a very important dog. This dog has a good heart.', 'Fire department and animal control personnel rescue a very scared and cold dog'] 17 tweets

4.0.3 [Optional! (will not be graded)] Question 8: trying to improve the model

- [2 points] a. Think of some ways to improve the scikit-learn Naive Bayes model by playing with the settings or applying linguistic preprocessing (e.g., by filtering on part-of-speech, or removing punctuation). Do not change the classifier but continue using the Naive Bayes classifier. Explain what the effects might be of these other settings
- [1 point] b. Apply the model with at least one new setting (train on the airline tweets using 80% training, 20% test) and generate the scores
- [1 point] c. Discuss whether the model achieved what you expected.

4.1 End of this notebook

```
[134]: # Preprocessing function (POS filtering: keep only nouns, verbs, adjectives)
def preprocess_text(text):
    doc = nlp(text.lower()) # Convert to lowercase
    tokens = [token.text for token in doc if token.pos_ in {"NOUN", "VERB",
    ↪ "ADJ"}]
    return " ".join(tokens)

# Preprocess tweets (apply POS filtering only)
airline_tweets_train.data = [preprocess_text(tweet) for tweet in
    ↪ airline_tweets_train.data]

# Split data into training (80%) and testing (20%)
X_train, X_test, y_train, y_test = train_test_split(
    airline_tweets_train.data, airline_tweets_train.target, test_size=0.2,
    ↪ random_state=42)

# Bag of Words Representation with min_df=2
vectorizer = CountVectorizer(min_df=2)
X_train_vec = vectorizer.fit_transform(X_train)
X_test_vec = vectorizer.transform(X_test)

# Train Naive Bayes Model
model = MultinomialNB()
model.fit(X_train_vec, y_train)

# Evaluate on test set
y_pred = model.predict(X_test_vec)
```

```

print("Classification Report on POS-Filtered Model:")
print(classification_report(y_test, y_pred, target_names=airline_tweets_train.
    ↪target_names))

# Discussion
print("\nDiscussion:")
print("- POS filtering keeps only nouns, verbs, and adjectives, removing_
    ↪unimportant words.")
print("- This may improve sentiment classification by focusing on meaningful_
    ↪words.")
print("- However, removing other words (e.g., adverbs, pronouns) could affect_
    ↪certain tweet contexts.")

```

```

-----
ExtraData                                Traceback (most recent call last)
Cell In[134], line 8
      5     return " ".join(tokens)
      7 # Preprocess tweets (apply POS filtering only)
----> 8 airline_tweets_train.data = [preprocess_text(tweet) for tweet in_
    ↪airline_tweets_train.data]
      10 # Split data into training (80%) and testing (20%)
      11 X_train, X_test, y_train, y_test = train_test_split(
      12     airline_tweets_train.data, airline_tweets_train.target, test_size=0
    ↪2, random_state=42)

Cell In[134], line 3, in preprocess_text(text)
      2 def preprocess_text(text):
----> 3     doc = nlp(text.lower()) # Convert to lowercase
      4     tokens = [token.text for token in doc if token.pos_ in {"NOUN",_
    ↪"VERB", "ADJ"}]
      5     return " ".join(tokens)

File ~/Desktop/anaconda3/envs/Text_Mining_python/lib/python3.12/site-packages/
    ↪spacy/language.py:1040, in Language.__call__(self, text, disable,_
    ↪component_cfg)
    1019 def __call__(
    1020     self,
    1021     text: Union[str, Doc],
    (...)
    1024     component_cfg: Optional[Dict[str, Dict[str, Any]]] = None,
    1025 ) -> Doc:
    1026     """Apply the pipeline to some text. The text can span multiple_
    ↪sentences,
    1027     and can contain arbitrary whitespace. Alignment into the original_
    ↪string
    1028     is preserved.
    (...)

```



```

1038     DOCS: https://spacy.io/api/language#call
1039     """
-> 1040     doc = self._ensure_doc(text)
1041     if component_cfg is None:
1042         component_cfg = {}

File ~/Desktop/anaconda3/envs/Text_Mining_python/lib/python3.12/site-packages/
↳spacy/language.py:1133, in Language._ensure_doc(self, doc_like)
    1131     return self.make_doc(doc_like)
    1132 if isinstance(doc_like, bytes):
-> 1133     return Doc(self.vocab).from_bytes(doc_like)
    1134 raise ValueError(Errors.E1041.format(type=type(doc_like)))

File ~/Desktop/anaconda3/envs/Text_Mining_python/lib/python3.12/site-packages/
↳spacy/tokens/doc.pyx:1362, in spacy.tokens.doc.Doc.from_bytes()

File ~/Desktop/anaconda3/envs/Text_Mining_python/lib/python3.12/site-packages/
↳srsly/msgpack_api.py:27, in msgpack_loads(data, use_list)
    25 # msgpack-python docs suggest disabling gc before unpacking large
↳messages
    26 gc.disable()
---> 27 msg = msgpack.loads(data, raw=False, use_list=use_list)
    28 gc.enable()
    29 return msg

File ~/Desktop/anaconda3/envs/Text_Mining_python/lib/python3.12/site-packages/
↳srsly/msgpack/_init_.py:85, in unpackb(packed, **kwargs)
    83     object_hook = functools.partial(decoder, chain=object_hook)
    84     kwargs["object_hook"] = object_hook
---> 85 return _unpackb(packed, **kwargs)

File ~/Desktop/anaconda3/envs/Text_Mining_python/lib/python3.12/site-packages/
↳srsly/msgpack/_unpacker.pyx:213, in srsly.msgpack._unpacker.unpackb()

ExtraData: unpack(b) received extra data.

```